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Development of an Artificial Intelligence (AI) Action Plan

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Red Hat AI in the Market: Products, Benefits, and Operationalization

Executive Summary

As part of the Office of Science and Technology Policy (OSTP) directive from Executive Order 14179 the government requires information on how to best move forward on the highest priority policy actions in the new AI Action Plan. After careful analysis of your request, Red Hat understands the information you require as falling into three primary categories: acquisition, security and management. OSTP needs to **acquire** the solution, its hardware, chips, datacenters, energy consumption - which creates a clear path to procurement. Capabilities for cybersecurity, national security and defense, intellectual property, data privacy and security throughout the lifecycle of artificial intelligence (AI) system development and deployment (to include security against AI model attacks), open source development, risks, explainability and assurance of AI model outputs, export controls are all components of a secure solution. Finally, it is essential to **manage** the solution by educating the workforce, fostering international collaboration, and establishing governance models for development, innovation, competition, and regulation. Building a robust, **secure**, and sustainable AI infrastructure is crucial to driving innovation that aligns with the needs of a decentralized government, even in edge environments.

To advance America's AI leadership, Red Hat recommends an AI Action Plan that is based on a Red Hat platform-centric approach. Our approach leverages the Red Hat AI product portfolio, which we describe with a summarization of benefits that the National Science Foundation (NSF) will realize with Red Hat AI solutions on an open architecture. Then, we move into an alignment of NSFs relevant AI policy topics organized in "acquire", "secure" and "manage" focus areas of the future plan.

As organizations begin implementing AI on-premises, the first step is often to assess whether their data centers are prepared. Upgrading IT infrastructure involves ensuring sufficient power and cooling, preparing the network to handle large data volumes, optimizing and expanding capacity, and implementing security measures while enabling scalability.

It's helpful to consider how automation plays a role in several key areas:

- ➢ Providing reliable, 100% uptime for AI workloads
- ➢ Ensuring networks, storage, data, and applications are seamlessly integrated and fully optimized
- ➢ Establishing teams and processes to manage load balancers, GPUs, and the flow of data to and from the edge

All of these elements of the infrastructure are critical to a successful AI implementation and ongoing strategy. Ansible Automation Platform plays a critical role in establishing a solid foundation for AI implementations by simplifying the deployment, configuration, and management of AI infrastructure components. We'll share specific examples spanning orchestration, operationalization, and infrastructure optimization.

We understand the criticality of cross operating platform solutions and the challenge to manage fiscal responsibility for tax payors. Red Hat helps our customers make sure they deliver the AI where their data is, and where they want it to be located, whether on-premises, in the cloud, or out at the edge - essentially across hybrid platforms for enabling federal surveillance.

Our products are not constrained by the "moats" of AI providers. These moats—proprietary hardware, models, exclusive datasets, and sheer scale—can make competing or achieving comparable results difficult.

Red Hat's AI stack is built on open, secure, and supported software, allowing customers to flexibly navigate these moats. Our open-source approach ensures compatibility with various hardware accelerators, from market leader NVIDIA to NPUs (Neural Processing Units) and even legacy consumer GPUs. Red Hat's platform runs across most, if not all, hardware stacks while seamlessly integrating with various software solutions, including but not limited to—vector databases, developer toolchains, open models from sources like Hugging Face, or compliance frameworks and more.

Regardless of the underlying hardware, the federal government can deploy and run any open-source AI model using Red Hat's AI solutions.

The capability that RedHat AI inference can bring to customers:

- Red Hat's recent acquisition of Neural Magic enables the integration of advanced software-accelerated AI inference, reducing the dependency on proprietary accelerators and models. This helps with cost-optimization, increased compatibility, and transparency into the open-source tool stack.
- Cost-effective, scalable AI solutions that enable organizations to quickly deploy high-performance AI models on standard CPU infrastructure using Red Hat's open-source platform for seamless integration, automation, and security, while reducing the need for expensive GPUs and minimizing infrastructure overhead.
- By maintaining data sovereignty (ownership of your data), reducing reliance on proprietary hardware and software, and utilizing open-source platforms, an enhanced security posture can be achieved while having efficient and high-performance AI inference capabilities on trusted open-source platforms, ensuring compliance across hybrid and edge environments.
- Access to a vast ecosystem of certifications, partners, and integrations supported on and with the Red Hat stack.

A "do it yourself" (DIY) approach is isolated and inconsistent. It's usually more costly, more complex, and less efficient. Planning, from the platform-up, provides a trusted, comprehensive, and consistent solution, giving more minutes back to the federal agency's mission. Trusted by more than 90% of the Fortune 500 today[1] Red Hat is helping businesses unlock new healthcare cures, enabling a future of autonomous vehicles, and opening paths to bring new banking services to market faster with constant innovation. In Figure 1, we end with a high level diagram of Red Hat's open hybrid cloud platform centric approach, as we move into the AI market challenges discussion.

AI Market Challenges and Operationalization

Organizations today face significant challenges in operationalizing AI. According to 2023 Gartner research, half of organizations require 7-12 months to operationalize an AI model from concept to limited production, with more than a quarter taking over a year to fully realize their AI initiatives.[2] This delay stems from multiple factors that impact implementation timelines and effectiveness.

Balancing innovation with cost management during AI implementation, as well as access to compute and GPUs has shown to be a primary challenge. **Skills shortages** present another major obstacle to modernizing applications with AI capabilities, as specialized expertise is difficult to source and retain. Enterprise AI deployments require **scalable, flexible, and consistent platforms** that reduce complexity while integrating with existing systems, yet building this infrastructure is often time-consuming and resource-intensive.

Data sovereignty and privacy concerns significantly impact how organizations can leverage data for AI implementations, creating compliance challenges that must be addressed. Organizations must build robust IT infrastructure to support AI/ machine learning (ML) experiments and implementations, while supporting rapid iteration, requiring significant investment in both hardware and expertise. Traditional approaches to model fine-tuning require specialized data science knowledge that is often in short supply, making it difficult to customize models for specific business needs.

Large proprietary models in the generative AI space present particular challenges, as they are expensive to run and difficult to train and tune for specific use cases. Organizations struggle to align these models to business requirements due to **complex training processes and high costs.** Successfully implementing AI requires clear ethics guidelines, compliance frameworks, and monitoring tools to prevent unintended consequences, adding another layer of complexity to already challenging projects.

Red Hat AI Product Portfolio Overview

Red Hat provides a comprehensive AI solution stack designed to address organizational needs across different AI maturity levels. This integrated portfolio enables a clear progression from initial exploration to production-scale implementation, while maintaining enterprise standards throughout the journey.

[1] https://www.redhat.com/en/about/company
[2] https://www.redhat.com/rhdc/managed-files/cl-gartner-pulse-survey-infographic-analyst-material-313472-202305-en.pdf

At the foundation level, Red Hat Enterprise Linux AI serves as a platform for initial AI deployments and entry-level use cases. Building on this foundation, Red Hat OpenShift AI delivers an end-to-end MLOps platform for enterprise-scale AI/ML model development and deployment. For model customization, InstructLab offers alignment and enhancement capabilities that make it possible to tailor large language models to specific domains. At the core of these solutions are Granite Language Models, which provide open source, Apache 2.0 licensed language and code models that are fully indemnified by Red Hat.

Unlike proprietary solutions, Red Hat's approach combines open source flexibility with enterprise-grade reliability, security, and support. Red Hat AI technologies build upon our established enterprise Linux and container platforms to ensure stability and compatibility, while integrating with and complementing existing investments in Red Hat technologies. The solutions include 24x7 enterprise support via a flexible subscription model and benefit from Red Hat's partner ecosystem, to enhance the core offerings with more capabilities and integration options.

Red Hat AI solutions support integration with popular open source AI frameworks and tools to maximize flexibility, while our product roadmap ensures continuous evolution of AI capabilities to address emerging requirements. These technologies benefit from community innovation and enterprise-focused development approaches, creating a balance between cutting-edge features and production stability. Perhaps most importantly, Red Hat AI solutions are designed to work seamlessly in hybrid environments spanning on-premises, public cloud, and edge deployments, providing the flexibility organizations need to address varying requirements.

Red Hat Enterprise Linux AI

Red Hat Enterprise Linux AI integrates enterprise-ready versions of InstructLab, Granite language models, and a bootable image of Red Hat Enterprise Linux into a comprehensive foundation for AI implementation. This integrated solution delivers pre-configured drivers for GPU acceleration and hardware optimization to accelerate deployment and time-to-value. The included Granite language models (7b) and code models (3b, 8b, 20b, 34b) provide cost and performance-optimized AI capabilities that serve as the core of enterprise AI applications.

Red Hat Enterprise Linux AI includes InstructLab model alignment tooling that enables efficient contribution of domain-specific knowledge to AI models, making customization accessible to a broader range of stakeholders. Full enterprise support and indemnification for running open-source large language models (LLMs), in production environments provides the confidence organizations need to deploy these models in business-critical applications. Red Hat Enterprise Linux AI serves as an ideal entry point for organizations beginning their AI journey with first proof-of-concept implementations, particularly for Retrieval-Augmented Generation (RAG) use cases that require cost-effective alternatives to expensive proprietary models.

By simplifying deployment across hybrid infrastructure environments through a complete, integrated solution stack, Red Hat Enterprise Linux AI helps organizations accelerate the process of going from proof-of-concept to production server-based deployments. The solution provides all necessary tools for training, tuning, and deploying models where data resides, anywhere across the hybrid cloud. Red Hat Enterprise Linux AI forms a foundation model platform for bringing open source-licensed generative AI models into the enterprise, lowering costs and removing barriers to testing and experimentation with generative AI technologies.

When organizations are ready to scale their AI initiatives, Red Hat Enterprise Linux AI provides an on-ramp to Red Hat OpenShift AI while maintaining the same tools and approaches. This continuity allows teams to leverage existing knowledge as they move to more sophisticated implementations. Red Hat Enterprise Linux AI streamlines the alignment of generative models to business requirements through accessible tooling, enabling organizations to realize value from AI more quickly while maintaining enterprise standards.

Red Hat OpenShift AI

Red Hat OpenShift AI delivers a comprehensive MLOps and LLMOps platform built on the OpenShift container orchestration system. This enterprise-scale solution provides self-service Jupyter notebooks and data science tools that empower data scientists and developers to work efficiently with their preferred tools. Red Hat OpenShift AI offers GPU-accelerated computing with automatic scaling capabilities to optimize resource utilization, a critical factor in managing the high costs associated with AI infrastructure.

Through streamlined operations, Red Hat OpenShift AI simplifies AI deployment at enterprise scale across multiple environments while facilitating multi-team collaboration across data science, engineering, and operations teams through a consistent user experience. The platform delivers hybrid cloud flexibility that allows deployment across on-premises, cloud, and edge environments to meet varying requirements. Red Hat OpenShift AI's automatic scaling capabilities represent a significant advantage, enabling GPU resources to scale down to zero when not in use, to improve cost efficiency, while supporting techniques like GPU sharding and co-location of various models within GPUs for maximized resource utilization.

Red Hat OpenShift AI helps organizations realize AI value while encouraging innovation and balancing costs, addressing one of the key challenges in AI adoption. The platform enables organizations to build, train, deploy, and monitor AI/ML workloads in on-premise data centers or close to where data is located, allowing businesses to evolve their AI strategy by moving operations to the cloud and edge when required. By integrating both open source communities and Red Hat's AI partner ecosystem, Red Hat OpenShift AI increases flexibility and freedom of choice when selecting AI/ML technologies.

The solution reduces the complexities of building and delivering models and AI-enabled applications that are accurate, reliable, and trustworthy. This comprehensive approach includes intensive computation power such as GPUs and streamlined operations to simplify AI solution delivery consistently and at scale across hybrid environments, including support for air-gapped and disconnected deployments in regulated industries.

InstructLab and Granite Language Models

InstructLab democratizes model customization by making it accessible to stakeholders beyond data scientists, addressing a significant barrier to AI adoption. The approach employs taxonomy-driven data curation using a hierarchical structure that organizes skills and knowledge areas, providing a structured framework for model enhancement. InstructLab's accessible contribution format uses simple human-readable data serialization format files, in question-and-answer pair format that domain experts without data science backgrounds can contribute to, enabling broader participation in model development.

The platform uses large-scale synthetic data generation through Skills Synthetic Data Generator (Skills-SDG) and Knowledge-SDG to efficiently expand training datasets. Skills-SDG uses prompt templates for instruction generation, evaluation, response generation, and final pair evaluation, while Knowledge-SDG generates instruction data for domains not covered by the teacher model using external knowledge sources. Our approach reduces a need for large amounts of manually annotated data, making model customization more efficient and cost-effective.

InstructLab implements a multi-phase training process with knowledge tuning to integrate factual information and skills tuning to enhance application capabilities. This multiphase approach helps maintain stability and prevents catastrophic forgetting through a replay buffer, ensuring that models retain previously learned capabilities while adding new ones. The result - continuous improvement without degradation often seen in traditional fine-tuning approaches.

Granite language models form the core AI capability within Red Hat's AI solutions, with the Granite 7b English language model and Granite 3b, 8b, 20b, and 34b code models addressing various use cases. These models are released under Apache 2.0 licensing, providing full transparency regarding datasets used for training. Red Hat provides complete enterprise indemnification for Granite models, assuming legal responsibility for the models' outputs and providing organizations with confidence in their deployment. The Granite models deliver optimized performance as cost-effective alternatives to larger proprietary models, enabling organizations to implement AI solutions with predictable costs and performance characteristics.

Benefits of Red Hat's AI Solutions

Red Hat's AI solutions deliver significant benefits through simplified adoption, reduced complexity in building and delivering AI-enabled applications, and integration of open source communities with Red Hat's AI partner ecosystem. This integrated approach provides maximum flexibility and innovation while increasing freedom of choice in AI/ML technologies without sacrificing enterprise support standards. The solutions deliver operational consistency through a unified user experience across data science, engineering, development, and operations roles, enabling effective collaboration across functions.

Self-service access to collaboration workflows and computing resources accelerates development cycles by removing bottlenecks between teams. Streamlined operations simplify delivery of AI solutions consistently and at scale, while integrated data science workflows within existing DevOps pipelines reduce friction between development and operations. Red Hat's solutions allow organizations to build, train, and deploy AI workloads where data is located to address regulatory requirements, providing flexibility to evolve AI strategy by seamlessly moving between on-premises, cloud, and edge environments as needs change.

For highly sensitive or regulated workloads, Red Hat supports air-gapped and disconnected environments with the same tools and capabilities available in connected deployments. Solutions accelerate time-to-market by removing common implementation barriers, while the enterprise subscription model provides customers with proven 24x7 support for AI initiatives. Red Hat's open-source approach avoids vendor lock-in while ensuring enterprise-grade reliability and security, so organizations balance innovation with cost management via resource optimization.

Perhaps most importantly, Red Hat's multi-level product portfolio allows NSF to start small and scale as AI initiatives mature, providing a clear growth path without requiring extensive reimplementation as needs evolve. This approach aligns with how organizations typically adopt AI, starting with focused use cases and expanding as they gain experience and confidence.

Open Radio Access Network

AI is well suited to benefit from the deployment flexibility of the distributed nature of service provider networks as well as helping to drive their operational efficiencies. Open source is working to help create bite-sized AI models, which can help lower cost and compute requirements while powering up an organization's existing applications and processes. And by leveraging these smaller AI models in the radio access network (RAN), core and more, service providers will be able to tackle a growing number of business and operational challenges.

By way of example: Red Hat, Inc., the world's leading provider of open source solutions, and SoftBank Corp. (“SoftBank”) today [March 3, 2025] announced the implementation of Artificial Intelligence Radio Access Network (AI-RAN) to optimize power consumption and networking performance using Red Hat OpenShift, the industry’s leading hybrid cloud application platform powered by Kubernetes. With this collaboration, Red Hat and SoftBank are addressing many of the long-standing RAN implementation challenges that service providers often face, such as balancing user demands with energy costs, resource availability and managing deterministic and distributed workloads.[3]

Acquire

Hardware and Computational Efficiency

Red Hat's AI solutions are designed to optimize hardware utilization across the AI lifecycle, from development through production deployment. The platform provides optimized drivers and configurations for leading GPU manufacturers including NVIDIA, AMD, and Intel, ensuring maximum performance from specialized AI acceleration hardware. This hardware optimization extends beyond GPUs to include support for emerging AI-specific chips and specialized processors that accelerate specific AI workloads.

OpenShift AI addresses the critical challenge of GPU resource management through intelligent scheduling and automatic scaling capabilities. The platform can automatically scale GPU resources up as demand increases and down to zero when not in use, significantly improving utilization rates and cost efficiency. Advanced techniques like GPU sharing, model co-location, and resource partitioning further enhance efficiency by allowing multiple workloads to share expensive computational resources without interference.

[3] https://www.redhat.com/en/about/press-releases/red-hat-and-softbank-corp-implement-ai-ran-optimize-network-performance-and-sustainability

For data center efficiency, Red Hat's solutions employ sophisticated workload placement strategies that consider factors like power consumption, cooling requirements, and overall data center utilization. Integration with data center management tools enables power-aware scheduling that can adjust computational intensity based on overall energy consumption patterns or time-of-day pricing models. The platform provides detailed monitoring of resource utilization and energy consumption, enabling organizations to identify optimization opportunities and track efficiency improvements over time.

Red Hat's AI infrastructure solutions incorporate leading practices for model optimization, including model quantization, pruning, and compression techniques that reduce computational requirements without significant performance degradation. Support for optimized inference engines like ONNX Runtime and TensorRT enables deployment of models tuned for production efficiency rather than development flexibility. These optimization capabilities are particularly important for edge deployments where power and computational resources may be constrained.

The demand for multi-vendor networks built on open principles is surging, fuelled by ongoing investment and predictions of surging growth. Open RAN – which hinges on interoperable interfaces and the disaggregation of hardware and software – has become a crucial avenue for meeting this demand. Ericsson, in collaboration with Dell Technologies and Red Hat, has been developing O2 interface definitions and capabilities. This interface facilitates communication between the Service Management and Orchestration (SMO) Framework and the O-Cloud Infrastructure Management Framework. For additional information regarding our partnerships: https://www.redhat.com/en/products/ai

AI Standards, Interoperability, and Procurement

Red Hat actively participates in and contributes to emerging AI standards initiatives, ensuring that its solutions remain compatible with evolving industry frameworks. The company's AI platforms support major interoperability standards like ONNX for model exchange, MLflow for experiment tracking, and Kubeflow for workflow orchestration. This standards-based approach enables integration with diverse toolsets and prevents vendor lock-in, providing organizations with flexibility as their AI strategies evolve.

For government procurement and compliance requirements, Red Hat offers solutions that align with frameworks like FedRAMP, NIST AI Risk Management, and emerging AI-specific certification programs. Our AI platforms can be configured to meet specific government security requirements, including those for classified environments and sensitive applications. Documentation and validation processes support compliance verification and audit requirements.

Red Hat's AI solutions are designed with interoperability as a core principle, supporting integration with existing enterprise systems and specialized AI tools from multiple vendors. The platform provides standard APIs and interfaces that enable connection to data sources, model repositories, and operational systems without requiring custom development. This interoperability extends to support for multiple AI frameworks and libraries, enabling organizations to use the most appropriate tools for specific use cases.

For procurement teams, Red Hat offers flexible licensing models that align with both traditional enterprise procurement processes and emerging consumption-based approaches. The subscription model provides predictable costs while ensuring access to the latest capabilities and security updates. Specialized licensing options support air-gapped environments, government applications, and educational institutions with specific requirements.

Red Hat's commitment to open standards and interoperability ensures that investments in AI infrastructure and applications remain viable as technologies evolve. The platform's modular architecture allows components to be updated or replaced individually as requirements change, protecting overall investment while enabling adoption of emerging capabilities. This approach is particularly valuable for long-term government projects where sustainability and maintainability are critical considerations.

Implementation Path and Maturity Model

Red Hat provides a clear progression path for organizations at different AI maturity levels, allowing them to start where they are and evolve as their capabilities and requirements grow. Organizations can begin with Red Hat Enterprise Linux AI for first RAG use cases, single-server deployments, and initial proofs of concept that build board-level visibility into AI initiatives. Red Hat Enterprise Linux AI is particularly well-positioned as the entry point for organizations experimenting with proprietary models but need more cost-effective solutions for production.

As deployments grow, Red Hat OpenShift AI enables scaling from individual servers to clusters with optimized resource management, facilitating team collaboration across larger projects with varied deployment requirements. Organizations with advanced needs can integrate with partner solutions like Watsonx for complex multi-model orchestration, creating a complete ecosystem for enterprise AI. This maturity model allows starting with accessible entry points while providing a clear growth path for expanding AI initiatives, with organizations leveraging Red Hat's expertise for guidance through each stage of the implementation journey.

Red Hat's solution progression allows organizations to start with one success point and gradually expand to multiple models and use cases without major reimplementation. The transition from Red Hat Enterprise Linux AI to OpenShift AI represents a natural evolution as organizations scale their AI implementations, supporting both horizontal scaling (more models) and vertical scaling (more complex models). This approach aligns with typical enterprise AI adoption patterns, from experimentation to enterprise-wide deployment, with the Red Hat solution stack evolving alongside NSF's capabilities and requirements to ensure continued alignment.

The progression path helps organizations avoid common pitfalls in scaling AI from proof-of-concept to production, providing a tested approach that reduces risk and accelerates value realization. This structured yet flexible path enables organizations to grow their AI capabilities at a pace that matches their business needs and readiness, without creating technical debt or implementation barriers as they scale.

Future-Proofing AI Investments with Red Hat

Red Hat's approach to AI helps organizations future-proof their investments through several key strategies. The open-source foundation provides access to the latest innovations from both community and commercial development, ensuring solutions remain current with evolving best practices. The modular architecture allows components to evolve independently as technology advances, preventing wholesale replacements as individual technologies mature. Red Hat's subscription model ensures continuous updates and improvements without disruptive version changes, providing a stable yet evolving platform.

Organizations can adapt their Red Hat AI implementations as business requirements evolve without major reimplementation, protecting initial investments while enabling growth. The hybrid cloud approach ensures flexibility to adapt deployment models as organizational needs change, preventing lock-in to specific infrastructure choices. Standardized APIs and interfaces support integration with emerging technologies, allowing organizations to incorporate new capabilities as they become available.

Red Hat's commitment to upstream open-source projects ensures alignment with industry-wide innovation, while consistent tools and approaches enable scaling from proof-of-concept to enterprise-wide deployment. Regular driver updates maintain compatibility with evolving hardware acceleration technologies, allow organizations to take advantage of performance improvements without application changes. The partner ecosystem provides access to complementary technologies to enhance AI investments, extending capabilities beyond Red Hat's core offerings.

Standards-based implementation ensures long-term compatibility and interoperability, while Red Hat's regular release cadence provides predictable enhancement paths for AI capabilities. Organizations can protect their AI investments through Red Hat's transparent roadmap and commitment to backward compatibility. The community-driven approach ensures diverse input into future development directions, creating solutions that address a broad range of requirements rather than focusing on narrow use cases.

Through these approaches, Red Hat helps organizations build AI solutions that can evolve alongside business needs and technology advancements, providing a foundation that supports both current requirements and future growth. This long-term perspective is particularly valuable in AI, where the technology landscape continues to evolve rapidly and organizations need solutions that can adapt without requiring complete reimplementation.

Secure

Cybersecurity and AI Model Protection

Red Hat's AI solutions incorporate comprehensive security capabilities that protect models, data, and infrastructure throughout the AI lifecycle. The platform builds on Red Hat's established security framework, with features specifically designed to address AI-related threats and vulnerabilities. Security is implemented at multiple levels, from the underlying infrastructure through model deployment and monitoring.

At the infrastructure level, Red Hat provides hardened operating system configurations optimized for AI workloads, with regular security updates and vulnerability management. Secure container environments isolate workloads and enforce least-privilege access controls, preventing unauthorized access to models and data. Integration with enterprise identity and access management systems enables fine-grained control over who can access and modify AI assets.

For model security, Red Hat implements protections against common attack vectors like model poisoning, prompt injection, and adversarial examples. The platform includes capabilities for input validation and sanitization that help prevent manipulation of model behavior through carefully crafted inputs. Monitoring tools detect unusual patterns that might indicate attempted attacks, enabling rapid response to potential security incidents.

Data security is addressed through comprehensive data lifecycle management capabilities that protect sensitive information during training, validation, and inference. The platform supports data encryption both at rest and in transit, with configurable policies for data handling based on sensitivity classification. Integration with enterprise data governance frameworks ensures consistent application of security policies across all data used in AI processes.

Red Hat's solutions enable organizations to implement defense-in-depth strategies for AI systems, with multiple layers of protection against various types of attacks. The platform supports implementation of security best practices like model versioning, controlled deployments, and comprehensive audit trails that document all interactions with models and data. These security capabilities are particularly important for applications in national security, defense, and critical infrastructure, where protection against sophisticated attacks is essential.

Intellectual Property and Innovation Protection

Red Hat's approach to AI intellectual property balances open innovation with appropriate protections for both the company and its customers. The open source licensing of Granite models under Apache 2.0 provides clear rights and responsibilities for model usage, modification, and distribution. This transparent licensing reduces legal uncertainty while enabling broad adoption and community enhancement.

For organizations concerned about intellectual property risks when using AI models, Red Hat provides comprehensive indemnification that protects against claims of infringement. This protection covers both the models themselves and outputs generated by the models when used within specified parameters. The indemnification extends to patent, copyright, and trade secret claims, giving organizations confidence to deploy these models in production environments.

Red Hat's AI platforms include features for protecting organizational intellectual property during model training and use. Data lineage tracking documents what information is used in model development, while access controls prevent unauthorized exposure of sensitive data or proprietary knowledge. Organizations can implement policies that govern what information can be used for model training and how models can be deployed, ensuring alignment with overall intellectual property strategy.

For innovation protection, Red Hat provides mechanisms for organizations to safeguard their enhancements and customizations to AI models. While the base models are open source, proprietary extensions, custom training data, and specialized applications can be protected as needed. The platform supports hybrid approaches that combine open source foundations with proprietary elements where appropriate for business strategy.

Red Hat actively engages with evolving intellectual property frameworks for AI, ensuring that its solutions adapt to changing legal interpretations and precedents. The company's legal expertise in open source licensing provides valuable guidance for navigating the complex intersection of AI and intellectual property law. This forward-looking approach helps organizations implement AI with appropriate intellectual property management while minimizing legal risk.

Red Hat AI Technology Details

Red Hat's AI technologies provide a complete stack of capabilities designed to address the full lifecycle of AI implementation. RHEL AI is a foundation model platform that simplifies bringing open-source licensed generative AI models into enterprise environments, unlocking the power of efficient models with full enterprise support. The solution streamlines alignment of generative models to specific business requirements through accessible tooling, enabling organizations to train and deploy AI anywhere in hybrid environments to address diverse needs. OpenShift AI functions as an end-to-end MLOps platform for developing, training, and serving models at scale, providing self-service access to collaboration workflows and computational resources to accelerate development. The platform delivers consistent user experiences across technical roles to facilitate more effective teamwork, with automatic scaling capabilities that optimize GPU resource utilization. As load increases on models, OpenShift AI scales resources up to meet demand, then scales down to zero when not used to optimize expensive GPU resources.

InstructLab significantly reduces the need for large amounts of manually annotated data through synthetic data generation, with an accessible YAML format that makes it possible for subject matter experts to contribute domain knowledge without technical expertise. Granite models provide enterprise-grade, open-source language and code capabilities with full transparency and indemnification, giving NSF confidence in AI implementations. RHEL AI includes these highly performant, collaboratively developed models from the InstructLab community, with extended model lifecycle and model IP indemnification to ensure confidence in model outputs. The open approach avoids vendor lock-in while providing the reliability and support that enterprise organizations require, creating a balance between innovation and production stability that is often difficult to achieve with other approaches.

Manage

Open Source Model Development and Governance

Red Hat's approach to AI is fundamentally built on open source principles, from the operating system foundation through the application layer. The company's AI solutions leverage and contribute to leading open source AI projects, ensuring that innovations benefit the broader community while maintaining enterprise-grade reliability. This commitment to open source extends to Red Hat's Granite language models, which are released under Apache 2.0 licensing with full transparency regarding training data and methodologies.

The open source approach provides significant advantages for model governance and assurance. Transparent model development processes enable comprehensive review and validation, while community scrutiny helps identify potential issues before they impact production systems. Red Hat's InstructLab methodology combines the innovation advantages of open source with structured governance processes that ensure model quality and reliability. The taxonomy-driven approach to model enhancement provides a clear framework for documenting model capabilities and limitations, supporting responsible deployment.

Red Hat actively participates in and contributes to open source AI governance initiatives, helping to establish best practices and standards for responsible AI development. The company's AI solutions incorporate these governance principles through features like model cards that document model characteristics, intended use cases, and known limitations. Comprehensive lineage tracking maintains records of data sources, training methodologies, and validation procedures to support audit requirements and regulatory compliance.

For organizations developing their own models, Red Hat provides tools and frameworks that support responsible development practices without requiring specialized governance expertise. Templates for documenting model characteristics, validation processes, and deployment constraints make it easier to incorporate governance into the development workflow. Integration with common CI/CD pipelines enables automated validation against governance requirements during the build and deployment process, ensuring consistent application of policies. Case Study: Army, Project Linchpin

AI Explainability, Ethics, and Regulatory Compliance

Red Hat recognizes that explainability and assurance are critical requirements for enterprise AI adoption, particularly in regulated industries. The company's AI platforms incorporate capabilities for model interpretation and explanation that help organizations understand how models arrive at specific outputs. These explainability features are particularly important for addressing regulatory requirements in sectors like financial services, healthcare, and government. For large language models, Red Hat provides tools to analyze model behaviors and identify potential biases or problematic patterns. InstructLab's structured approach to model enhancement includes evaluation phases that assess model outputs against defined criteria for accuracy, fairness, and appropriateness. The platform supports integration with third-party evaluation frameworks that provide additional validation against industry-specific requirements.

Red Hat's AI solutions incorporate comprehensive monitoring capabilities that track model performance, drift, and potential issues in production environments. These monitoring tools help organizations identify when models are operating outside their intended parameters or when inputs vary significantly from training data. Automated alerting capabilities notify appropriate teams when potential issues arise, enabling proactive management of AI systems.

The platform supports implementation of ethics guidelines and responsible AI principles through configurable guardrails and validation processes. Organizations can define and enforce policies regarding data usage, model deployment, and application constraints through centralized governance controls. These capabilities are particularly important for government applications, where adherence to specific ethics guidelines and regulations may be mandatory.
Red Hat's solutions are designed to adapt to evolving regulatory requirements across jurisdictions, with flexible deployment options that support compliance with data sovereignty

and privacy regulations. The platform enables isolation of sensitive data and processing where required, while maintaining consistent management across environments. Regular updates ensure compatibility with emerging standards and regulations, providing organizations with confidence that their AI deployments will remain compliant as requirements evolve.

Workforce Development and Education

Red Hat recognizes that addressing AI skills gaps requires comprehensive education and workforce development strategies. The company provides extensive training and certification programs that help organizations build internal AI capabilities across technical and business roles. These education resources range from introductory concepts to advanced implementation techniques, supporting skills development at all levels.

For technical teams, Red Hat offers specialized training on AI infrastructure, model development, and operational management. These programs combine theoretical knowledge with hands-on experience using Red Hat's AI platforms, ensuring that participants develop practical skills that can be immediately applied. Certification pathways provide recognized validation of AI expertise, helping organizations identify qualified personnel for critical roles.

Red Hat supports broader workforce adaptation to AI through educational resources focused on responsible implementation and governance. These materials help business leaders, policy makers, and functional specialists understand AI capabilities, limitations, and implications without requiring deep technical knowledge. Customized education programs address the specific needs of different roles involved in AI initiatives, from executive sponsors to domain experts contributing to model development.

Through partnerships with educational institutions, Red Hat contributes to developing the next generation of AI practitioners. The company provides academic access to its AI platforms, curriculum materials, and technical expertise to support AI education in universities and technical schools. Internship and apprenticeship programs offer practical experience to emerging talent, helping bridge the gap between academic knowledge and industry requirements.
Red Hat's commitment to open source means much of its educational material is freely available to the broader community, supporting wider AI skills development. Red Hat actively participates in industry initiatives to establish skill standards and career pathways for AI roles, helping to create a more structured approach to workforce development in this rapidly evolving field.

International Collaboration and Responsible AI Development

Red Hat's global presence and open source foundation enable international collaboration on AI development and implementation. The company participates in multinational AI research initiatives, standards development, and policy frameworks that advance responsible AI practices across borders. This international engagement ensures that Red Hat's AI solutions incorporate diverse perspectives and address requirements across regions and regulatory environments.

The open nature of Red Hat's AI platforms facilitates collaboration between organizations, researchers, and developers worldwide. Shared models, methodologies, and best practices accelerate innovation while improving quality through diverse input and validation. International

collaboration is particularly valuable for addressing global challenges that require coordinated AI application, from climate change to public health.
Red Hat supports responsible AI development through incorporation of emerging global frameworks and principles into its platforms and practices. The company's solutions enable implementation of features like privacy-by-design, fairness validation, and appropriate human oversight that align with international guidelines for ethical AI development. These capabilities help organizations implement AI in ways that respect human rights, privacy, and cultural considerations across global operations.

For organizations working across multiple jurisdictions, Red Hat provides flexibility to adapt deployments to local requirements while maintaining consistent management and governance. The platform supports configuration of data handling, model deployment, and monitoring practices based on regional regulations and cultural expectations. This adaptability is particularly important for multinational enterprises and international agencies that must navigate complex compliance landscapes.

Red Hat actively engages with international policy development for AI governance, contributing technical expertise and implementation perspective to emerging frameworks. The company's practical experience with enterprise AI deployment informs realistic approaches to governance that balance innovation with appropriate controls. This engagement helps ensure that regulatory frameworks are technically feasible and effectively address legitimate concerns without unnecessarily constraining beneficial applications.

AI Customers

Figure 1. AI/ML customers

- [*U.S. Army, Project Linchpin*]
- [*Improving patient outcomes with Guidehouse, Red Hat, and AI*]
- [*Banco Galicia incorporates new business clients in a matter of minutes with the intelligent NLP platform*]